

MOMEN AREF

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ABOUT ME

Control and Automation Engineering student with hands-on experience in dynamic system modelling, control system design, and simulation using MATLAB/Simulink. Experienced in developing and evaluating control algorithms including PID and Model Predictive Control (MPC) for autonomous systems. Currently expanding my expertise in Model-Based Development (MBD), embedded systems, and control software implementation. Interested in automotive control systems, electrified powertrains, and embedded control engineering.

EDUCATION

Yildiz Technical University B.S. in Control and Automation Engineering | 2021-2026

- **Relevant Coursework:** Intelligent Control Systems, Embedded Systems, Model-Based Design (Currently Enrolled).

SKILLS

- **Technical:** Control Theory, Advanced Control Strategies (MPC, PID), Embedded Systems, Industrial Automation. Currently developing skills in Model-Based Development (MBD) using MATLAB/Simulink.
- **Software & Tools:** MATLAB/Simulink, C/C++, Python, STM32, Arduino.
- **Soft Skills:** Leadership, Problem Solving, Logical Thinking, Adaptability, Teamwork.
- **Languages:** Arabic (Native), English (Fluent), Turkish (Conversational).

PROJECTS

Two-Wheeled Self-Balancing Robot

- Designed and constructed a self-balancing robot utilizing an STM32 microcontroller.
- Utilized Simulink for system modelling and implemented a real-time PID control algorithm.

Path-Following Quadcopter

- Designed and simulated a path-following quadcopter using Model Predictive Control (MPC) in MATLAB.

Ball Balancing Robot

- Collaborated with a team to build a ball-balancing robot using a camera and computer vision to detect and control the ball's position.
- Responsible for designing and simulating the robot's control system.

WORK EXPERIENCE

NTG Clarity | Software Developer | *July 2023 - November 2023*

- Responsible for designing, building, and maintaining customized business applications and workflows for clients in various industries.

EFT | Warehouse Manager | *December 2021 - November 2022*

- Managed inventory and logistics for a warehouse, ensuring accurate stock levels and timely order fulfillment.
- Maintained and updated the inventory database system.

ACTIVITIES

University Student Union | President

- Served as President after progressing from Head of the Activities Committee to Vice President.
- Led planning, logistics, and communication efforts to enhance student engagement.
- Organized and coordinated multiple camps and student activities in Istanbul.